

Ejercicio No. 1 (50%) – Para las siguientes dos CFGs:

1. $S \rightarrow 0A0 \mid 1B1 \mid BB$

$A \rightarrow C$

$B \rightarrow S \mid A$

$C \rightarrow S \mid \varepsilon$

a) $C \rightarrow \varepsilon$ $A \rightarrow C \rightarrow \varepsilon$

$B \rightarrow A \rightarrow C \rightarrow \varepsilon$

$S \rightarrow 0A0 \mid 1B1 \mid BB$

↳ $S \rightarrow 0A0 \mid 0C0 \mid 1B1 \mid 1\varepsilon1 \mid BB \mid B$

$A \rightarrow C$

$B \rightarrow S \mid A$

$C \rightarrow S$

b) $S \rightarrow 0A0 \mid 0C0 \mid 1B1 \mid 1\varepsilon1 \mid BB \mid B$

$A \rightarrow 0A0 \mid 0C0 \mid 1B1 \mid 1\varepsilon1 \mid BB \mid B$

$B \rightarrow 0A0 \mid 0C0 \mid 1B1 \mid 1\varepsilon1 \mid BB \mid B$

c) ya se eliminaron los inútiles

$$d) \quad X_0 \rightarrow 0 \quad X_1 \rightarrow 1$$

$$0A0 \rightarrow X_0 A X_0$$

$$1B1 \rightarrow X_1 B X_1$$

$$0C \rightarrow X_0 X_0$$

$$11 \rightarrow X_1 X_1$$

$$Y \rightarrow A X_0$$

$$Z \rightarrow B X_1$$

$$X_0 A X_0 \rightarrow X_0 Y$$

$$X_1 B X_1 \rightarrow X_1 Z$$

$$S \rightarrow X_0 Y \mid X_0 X_0 \mid X_1 Z \mid X_1 X_1 \mid B B$$

$$A \rightarrow X_0 Y \mid X_0 X_0 \mid X_1 Z \mid X_1 X_1 \mid B B$$

$$B \rightarrow X_0 Y \mid X_0 X_0 \mid X_1 Z \mid X_1 X_1 \mid B B$$

$$Y \rightarrow A X_0$$

$$Z \rightarrow B X_1$$

$$X_0 \rightarrow 0$$

$$X_1 \rightarrow 1$$

$$2. \quad S \rightarrow aAa \mid bBb \mid \varepsilon$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid \varepsilon$$

$$D \rightarrow A \mid B \mid ab$$

a/ $S \rightarrow \varepsilon \quad C \rightarrow \varepsilon$

$$A \rightarrow C \rightarrow \varepsilon \quad B \rightarrow C \rightarrow \varepsilon$$

$$C \rightarrow CDE \rightarrow C \rightarrow \varepsilon$$

$$D \rightarrow A \rightarrow C \rightarrow \varepsilon$$

$$D \rightarrow B \rightarrow C \rightarrow \varepsilon$$

$$S \rightarrow aAa \rightarrow A \rightarrow C \rightarrow \varepsilon$$

$$S \rightarrow bBb \rightarrow B \rightarrow C \rightarrow \varepsilon$$

$$S \rightarrow aAa \mid bBb$$

$$\hookrightarrow aAa \mid aa \mid bBb \mid bb$$

$$A \rightarrow C \mid a$$

$$B \rightarrow C \mid b$$

$$C \rightarrow CDE \mid DE \mid CE \mid E$$

$$D \rightarrow A \mid B \mid ab$$

b/ $\hookrightarrow aAa \mid aa \mid bBb \mid bb$

$A \rightarrow a | CDE | DE | CE$

$B \rightarrow b | CDE | DE | CE$

$C \rightarrow CDE | DE | CE$

$D \rightarrow ab | a^1b | CDE | DE | CE$

c) $E \Rightarrow$ No productor

$S \rightarrow aAa | aabBb | bbb$

~~$A \rightarrow a | CDE | DE | CE$~~

~~$B \rightarrow b | CDE | DE | CE$~~

~~$C \rightarrow CDE | DE | CE$~~

~~$D \rightarrow ab | a^1b | CDE | DE | CE$~~

$C \Rightarrow$ se queda sin producciones $\Rightarrow$ No Productor

$S \rightarrow aAa | aabBb | bbb$

$A \rightarrow a |$

$B \rightarrow b |$

$$D \rightarrow ab \mid a \mid b$$

$D \Rightarrow$ no alcanzable se elimina

$$S \rightarrow a A a \mid a a \mid b B b \mid b b$$

$$A \rightarrow a \mid$$

$$B \rightarrow b \mid$$

d) $X_0 \rightarrow a \quad X_1 \rightarrow b$

$$a A a \rightarrow X_0 A X_0$$

$$b B b \rightarrow X_1 B X_1$$

$$a a \rightarrow X_0 X_0$$

$$b b \rightarrow X_1 X_1$$

$$\gamma \rightarrow A X_0$$

$$\gamma \rightarrow B X_1$$

$$X_0 A X_0 \rightarrow X_0 \gamma$$

$$X_1 B X_1 \rightarrow X_1 \gamma$$

$$c \Rightarrow \gamma \gamma \mid \gamma \gamma \mid \gamma \gamma \mid \gamma \gamma \mid \gamma \gamma$$

$S \rightarrow x_0 y \mid x_0 x_0 \mid x_1 y \mid x_1 x_1$

$Y \rightarrow AX_0$

$Z \rightarrow BX_1$

$A \rightarrow a$

$B \rightarrow b$

$x_0 \rightarrow a$

$x_1 \rightarrow b$